

CFD 2025-26 – Lab 09 – solution

The full code for this lab is provided in `CFDlab09.ipynb`: in Exercise 1, the modifications required by the different stabilization strategies are highlighted in the code.

Exercise 1

- 1 We define the data of the problem and complete the function `nonlinear_iteration_forms` with the forms `a_fp` and `L_fp`, which implement the Picard iteration as in Lab 7. Then, before entering the `for` loop of the iterative algorithm, we define a `LinearVariationalProblem` based on such forms and a corresponding `LinearVariationalSolver`. At each iteration, we solve a step of the method by simply calling the `solve` method of the solver: updating `u_old` by means of the `assign` method ensures that the whole variational problem is automatically updated and re-assembled.

Running the simulation, we can see that convergence (in terms of the stopping criterion based on the increment) is NOT attained: indeed, solving a fully Dirichlet problem entails an indetermination of the pressure field, as we have observed also in the convergence analysis of Lab02.

- 2 Even introducing the nullspace corresponding to constant modes in pressure (see `nsp` in the code) does not fix the issue: the linear solver is not converging. This is due to the choice of spaces $\mathbb{P}^1/\mathbb{P}^1$ is not an inf-sup stable finite element pair.

- 3 - 4 We introduce the Brezzi-Pitkäranta stabilization terms as additional forms constructed in the function `BrezziPitkaranta`, which takes as input also the piecewise-constant stabilization parameter δ_K , which has to be constructed. A piecewise constant `Function` is constructed by means of a `FunctionSpace` of type 'DG' of order 0. Then the value $h_K^2 \text{Re}$ is assigned to it through the `project` method. The variational problem is then build up using the sum of the original variational form/functional and the corresponding Brezzi-Pitkäranta terms, obtaining the result of Figure ?? - center. By solving the problem with different values of Re , we obtain the results reported in Figure 1. For small values of Re (3, 30), the solution is not much dissimilar from the solution of the Stokes problem, already commented in previous labs. For increasing values of Re (300), we notice that the cavity vortex is “larger” (in the sense that the region of non-negligible velocity has a wider diameter) and its pivotal point moves towards the center of the domain, driven by the increased inertial effect. Think about stirring tea in a mug: stronger stirring induces a stronger vortex, and its center is naturally driven towards the center of the mug.

In the case $\text{Re} = 3000$, however, the inertial effects are too strong: we find ourself in an advection-dominated regime and the standard fixed-point iteration method does not reach convergence in 100 iterations. Indeed, the solution clearly shows numerical instabilities in both the pressure and velocity fields

- 5 The SUPG stabilization provided in the text of the Exercise comes from the identification of the skew-symmetric part of the operator associated to the (fixed-point-)linearized

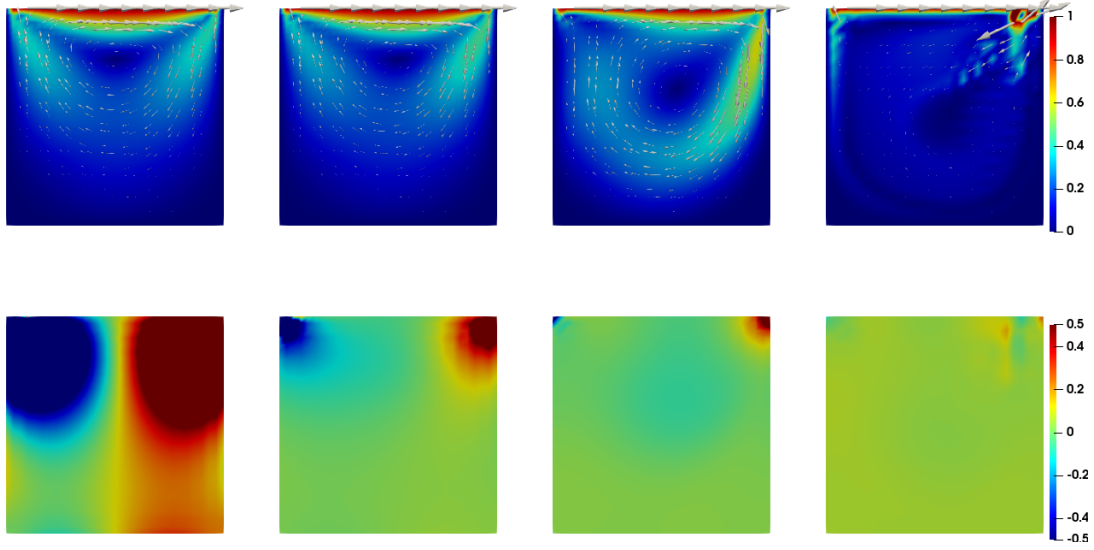


Figure 1: Velocity (above) and pressure (below) for $\text{Re}=3, 30, 300$ at convergence, and $\text{Re}=3000$ after 100 iterations, of the Brezzi-Pitkäranta-stabilized fixed-point iteration method.

problem. In particular, we point out that the operator

$$L_{\text{SS}}^{\text{adv}} \mathbf{v} = (\mathbf{u}_{k-1} \cdot \nabla) \mathbf{v} + \frac{1}{2} \text{div}(\mathbf{u}_{k-1}) \mathbf{v}$$

corresponds to the skew-symmetric part of the advection term for either choice of the advection form, namely

$$\begin{aligned} \text{either for } c(\mathbf{u}_{k-1}, \mathbf{u}, \mathbf{v}) &= ((\mathbf{u}_{k-1} \cdot \nabla) \mathbf{u}, \mathbf{v}) \\ \text{or for } \tilde{c}(\mathbf{u}_{k-1}, \mathbf{u}, \mathbf{v}) &= c(\mathbf{u}_{k-1}, \mathbf{u}, \mathbf{v}) + \left(\frac{1}{2} \text{div}(\mathbf{u}_{k-1}) \mathbf{u}, \mathbf{v} \right). \end{aligned}$$

Regarding the piecewise constant quantities $\text{Re}_K, \bar{u}_K, \delta_K$ in the case of the SUPG method, a comment is worth making about their implementation. We have seen that piecewise-constant functions belong to the $\mathbb{P}_0$ space, which is constructed in **Firedrake** by

```
FunctionSpace(mesh, 'DG', 0)
```

We observe now that the L^2 projection of a generic function z onto a FE space X_h corresponds to the solution of the following problem:

$$\text{Find } w_h \in X_h \text{ s.t. } \int_{\Omega} w_h \psi_h d\mathbf{x} = \int_{\Omega} z \psi_h d\mathbf{x} \quad \forall \psi_h \in X_h.$$

If X_h is the $\mathbb{P}_0$ space, the solution of this problem can be proven to be

$$w_h \in X_h \text{ s.t. } w_h|_K = \frac{1}{|K|} \int_K z d\mathbf{x} \quad \forall K \in \mathcal{T}_h.$$

Therefore, the element-wise averaged velocity magnitude $\bar{u}_K$ can be computed as follows:

```
ubar = Function(FunctionSpace(mesh, 'DG', 0))
ubar.project(sqrt(inner(u_old, u_old)))
```

and the stabilization parameter can be represented by a piecewise linear `Function` `delta_K` defined as follows:

```
delta_K.project(delta * conditional(
    gt(h * ubar * Re, 1.0),
    h/(ubar+1e-5),
    h*h*Re))
```

where the `conditional` ternary operator checks the condition encoded in the first input argument ($Re_K > 1$) and returns either the second or the third argument if the condition is true or false. Notice that we add a small perturbation of 10^{-5} in the denominator, to avoid possible division by 0.

We can define the SUPG terms by a function `supg_forms` returning a lhs form `lhs_supg` and a rhs functional `rhs_supg`, similarly to `nonlinear_iteration_forms`. Then, the fixed-point iteration `for` loop is modified to update the value of `delta_K`.

The results for $Re = 3000$, now stable, are reported in Figure 2. We can see that the spurious oscillations are completely removed from the velocity field, whose strong gradients are consistent with physics now. In the pressure field, some oscillations can still be seen near to the moving lid endpoints. This issue is not directly a result of the convection-dominated regime, but it is rather due to the coarseness of the mesh, that is not able to capture the strong pressure gradients in these region. Indeed, redistributing the mesh vertices in order to have a more refined mesh at the corners, while keeping the same number of dofs, removes these numerical oscillations. We refer directly to the code contained in `CFDLab09.ipynb` to see how the mesh is modified.

Exercise 2

- 1 We report the full solution to this point only for the case of $G(\mathbf{u}) = \varepsilon(\mathbf{u})$ and free-slip conditions.

Let us introduce the following trial and test spaces, encoding in the strong sense the Dirichlet conditions:

$$V = \{\mathbf{v} \in [H^1(\Omega)]^2 : \mathbf{v} = 6y(1-y)\mathbf{i} \text{ on } \Gamma_{\text{left}}, \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{\text{bottom}}, \mathbf{v} \cdot \mathbf{n} = 0 \text{ on } \Gamma_{\text{top}}\},$$

$$V_0 = \{\mathbf{v} \in [H^1(\Omega)]^2 : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{\text{left}} \cup \Gamma_{\text{bottom}}, \mathbf{v} \cdot \mathbf{n} = 0 \text{ on } \Gamma_{\text{top}}\}, \quad Q = L^2(\Omega),$$

and let $\mathcal{R}\mathbf{u}_L \in V$ be the lifting of the inflow velocity profile. Testing the momentum equation by a generic test function $\mathbf{v} \in V_0$ and using the integration by parts formula we

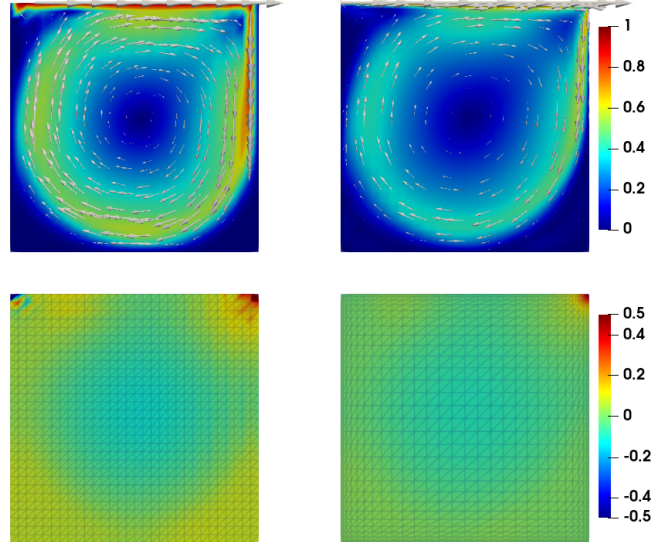


Figure 2: Velocity (above) and pressure (below) for $\text{Re} = 3000$ obtained with the SUPG-stabilized fixed-point iteration method. Left: uniform mesh; Right: non uniform mesh more refined at the corners (same number of elements as the uniform mesh).

obtain:

$$\begin{aligned}
 & \int_{\Omega} (\mathbf{u} \cdot \nabla) \mathbf{u} \cdot \mathbf{v} \, d\mathbf{x} - \frac{1}{\text{Re}} \int_{\Omega} \text{div} \, \varepsilon(\mathbf{u}) \cdot \mathbf{v} \, d\mathbf{x} + \int_{\Omega} \nabla p \cdot \mathbf{v} \, d\mathbf{x} = 0 \\
 & \quad \Updownarrow \\
 & \int_{\Omega} (\mathbf{u} \cdot \nabla) \mathbf{u} \cdot \mathbf{v} \, d\mathbf{x} + \frac{1}{\text{Re}} \int_{\Omega} \varepsilon(\mathbf{u}) \cdot \nabla \mathbf{v} \, d\mathbf{x} - \int_{\Omega} p \, \text{div} \, \mathbf{v} \, d\mathbf{x} - \int_{\partial\Omega} \left(\frac{1}{\text{Re}} \varepsilon(\mathbf{u}) - pI \right) \mathbf{n} \cdot \mathbf{v} \, d\mathbf{x} = 0 \\
 & \quad \Updownarrow \\
 & \int_{\Omega} (\mathbf{u} \cdot \nabla) \mathbf{u} \cdot \mathbf{v} \, d\mathbf{x} + \frac{1}{\text{Re}} \int_{\Omega} \varepsilon(\mathbf{u}) \cdot \nabla \mathbf{v} \, d\mathbf{x} - \int_{\Omega} p \, \text{div} \, \mathbf{v} \, d\mathbf{x} - \int_{\Gamma_{\text{top}}} \left(\frac{1}{\text{Re}} \varepsilon(\mathbf{u}) - pI \right) \mathbf{n} \cdot \mathbf{v} \, d\mathbf{x} = 0
 \end{aligned}$$

where the integrals over $\Gamma_{\text{left}} \cup \Gamma_{\text{bottom}}$ and Γ_{right} vanish because $\mathbf{v} = \mathbf{0}$ and $(\frac{1}{\text{Re}} \varepsilon(\mathbf{u}) - pI) \mathbf{n} = \mathbf{0}$, respectively, due to the boundary conditions. In order to account for the mixed free-slip condition on Γ_{top} , we resort to the decomposition of a vector function $\mathbf{v}$ along its normal and tangential component, namely $\mathbf{v} = (\mathbf{v} \cdot \mathbf{n}) \mathbf{n} + (\mathbf{v} \cdot \mathbf{t}) \mathbf{t}$. The integral over Γ_{top} can be then rewritten as follows:

$$\begin{aligned}
 & \int_{\Gamma_{\text{top}}} \left(\frac{1}{\text{Re}} \varepsilon(\mathbf{u}) - pI \right) \mathbf{n} \cdot [(\mathbf{v} \cdot \mathbf{n}) \mathbf{n} + (\mathbf{v} \cdot \mathbf{t}) \mathbf{t}] \, d\mathbf{x} = \\
 & \int_{\Gamma_{\text{top}}} \left(\frac{1}{\text{Re}} \varepsilon(\mathbf{u}) - pI \right) \mathbf{n} \cdot \mathbf{n} (\mathbf{v} \cdot \mathbf{n}) \, d\mathbf{x} + \int_{\Gamma_{\text{top}}} \left(\frac{1}{\text{Re}} \varepsilon(\mathbf{u}) - pI \right) \mathbf{n} \cdot \mathbf{t} (\mathbf{v} \cdot \mathbf{t}) \, d\mathbf{x} = 0,
 \end{aligned}$$

where the first integral vanishes due to the strong condition $\mathbf{v} \cdot \mathbf{n} = 0$, and the second one vanishes due to the no-friction condition $(\frac{1}{\text{Re}} \varepsilon(\mathbf{u}) - pI) \mathbf{n} \cdot \mathbf{t} = 0$.

Testing also the continuity equation, we can obtain the following weak formulation of the

problem: Find $\mathbf{u} = \check{\mathbf{u}} + \mathcal{R}\mathbf{u}_L$, with $\check{\mathbf{u}} \in V_0$, and $p \in Q$ s.t.

$$\begin{aligned} & \int_{\Omega} (\check{\mathbf{u}} \cdot \nabla) \check{\mathbf{u}} \cdot \mathbf{v} \, d\mathbf{x} + \frac{1}{\text{Re}} \int_{\Omega} \varepsilon(\check{\mathbf{u}}) \cdot \nabla \mathbf{v} \, d\mathbf{x} - \int_{\Omega} p \operatorname{div} \mathbf{v} \, d\mathbf{x} - \int_{\Omega} q \operatorname{div} \check{\mathbf{u}} \, d\mathbf{x} \\ &= - \int_{\Omega} (\check{\mathbf{u}} \cdot \nabla) \mathcal{R}\mathbf{u}_L \cdot \mathbf{v} \, d\mathbf{x} - \int_{\Omega} (\mathcal{R}\mathbf{u}_L \cdot \nabla) \check{\mathbf{u}} \cdot \mathbf{v} \, d\mathbf{x} - \int_{\Omega} (\mathcal{R}\mathbf{u}_L \cdot \nabla) \mathcal{R}\mathbf{u}_L \cdot \mathbf{v} \, d\mathbf{x} \\ & \quad - \frac{1}{\text{Re}} \int_{\Omega} \varepsilon(\check{\mathbf{u}}) \cdot \nabla \mathbf{v} \, d\mathbf{x} + \int_{\Omega} q \operatorname{div} \mathcal{R}\mathbf{u}_L \, d\mathbf{x} \quad \forall (\mathbf{v}, q) \in V_0 \times Q. \end{aligned}$$

Since the inner product between a skew-symmetric tensor and a symmetric one is equal to zero, and a generic tensor can be split into the sum of a symmetric and a skew-symmetric tensor ¹ the diffusion term can be equivalently written as

$$\frac{1}{\text{Re}} \int_{\Omega} \varepsilon(\mathbf{u}) \cdot \varepsilon(\mathbf{v}) \, d\mathbf{x}.$$

- 2 As it can be seen from the previous point, the differences between the different choices are in the diffusion term, which can read as one of the following

```
1.0/Re * inner(grad(uh), grad(v)) * dx      # G=grad(u)
1.0/Re * inner(sym(grad(uh)), grad(v)) * dx  # G=eps(u)
1.0/Re * inner(sym(grad(uh)), sym(grad(v))) * dx # G=eps, equivalent
```

and in the boundary conditions on Γ_{top} , which can be one of the following two

```
bc_noslip = DirichletBC(W.sub(0), Constant((0.,0.)), 4)
bc_freeslip = DirichletBC(W.sub(0).sub(1), Constant(0.), 4)
```

The results for all these choices are displayed in Figure 3, 4, 5. We can see that choosing free-slip conditions instead of no-slip ones, the wall velocity is different from zero, and the inflow parabolic profile transitions towards another (half-)parabolic profile with the maximum attained on Γ_{top} : such profile could be seen on a way longer pipe. Regarding the strain tensor, the choice affects the physical meaning of Neumann boundary conditions on Γ_{right} : if $G(\mathbf{u}) = \nabla \mathbf{u}$, the flow remains parallel to the walls and the outlet represents a fictitious interface between the domain and a continuation of the pipe, whereas if $G(\mathbf{u}) = \varepsilon(\mathbf{u})$, the outlet represents the end of the pipe in an open environment, with the flow having a component also in the transverse direction with respect to the walls.

Therefore, the flow between two infinite, rough planes can be modeled by Navier-Stokes equations with the symmetric strain tensor $\varepsilon(\mathbf{u})$ and no-slip boundary conditions.

¹For any $A \in \mathbb{R}^{n \times n}$, $A = \operatorname{sym}(A) + \operatorname{skw}(A)$, where $\operatorname{sym}(A) = \frac{A+A^T}{2}$ is symmetric and $\operatorname{skw}(A) = \frac{A-A^T}{2}$ is skew-symmetric.

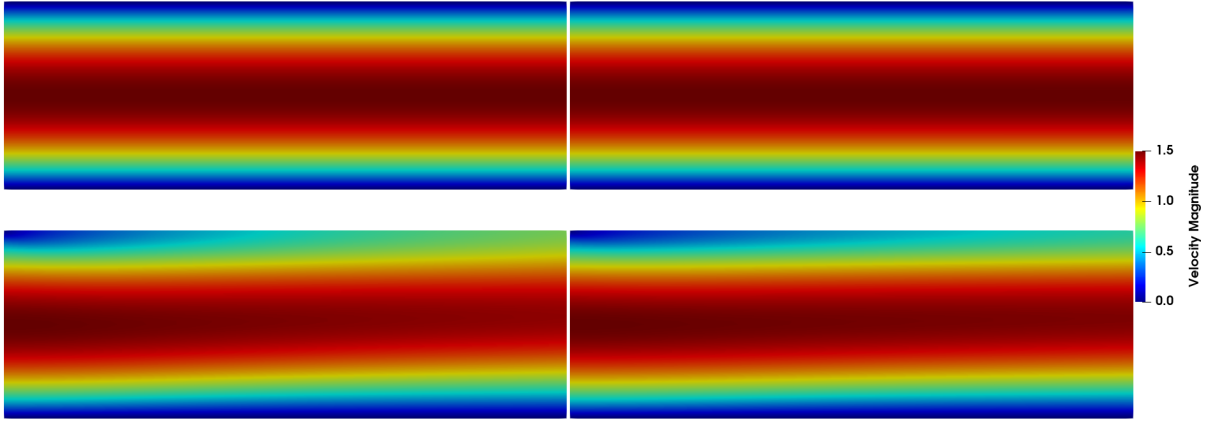


Figure 3: Exercise 3 - velocity. Left: $G(\mathbf{u}) = \nabla(\mathbf{u})$; right: $G(\mathbf{u}) = \varepsilon(\mathbf{u})$; top: no-slip on Γ_{top} ; bottom: free slip on Γ_{top} .

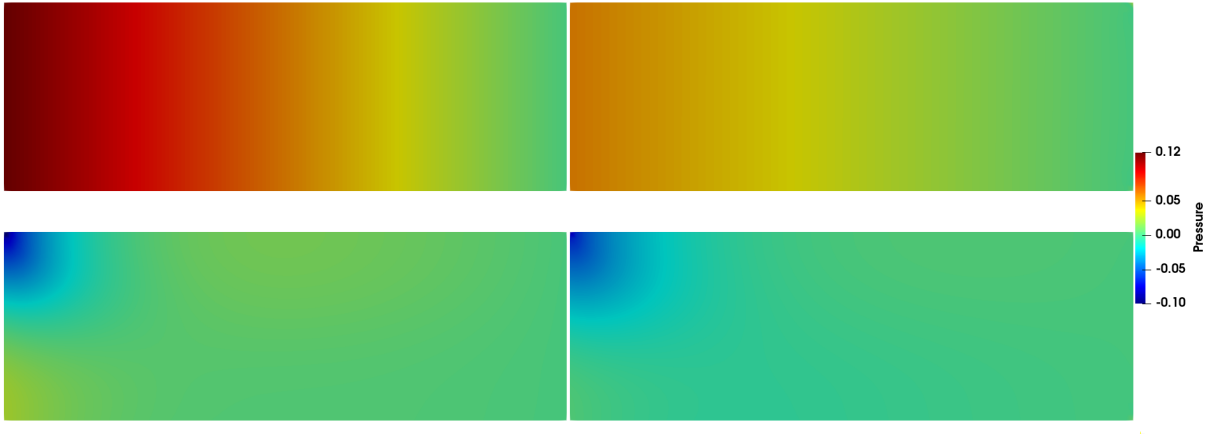


Figure 4: Exercise 3 - pressure. Left: $G(\mathbf{u}) = \nabla(\mathbf{u})$; right: $G(\mathbf{u}) = \varepsilon(\mathbf{u})$; top: no-slip on Γ_{top} ; bottom: free slip on Γ_{top} .

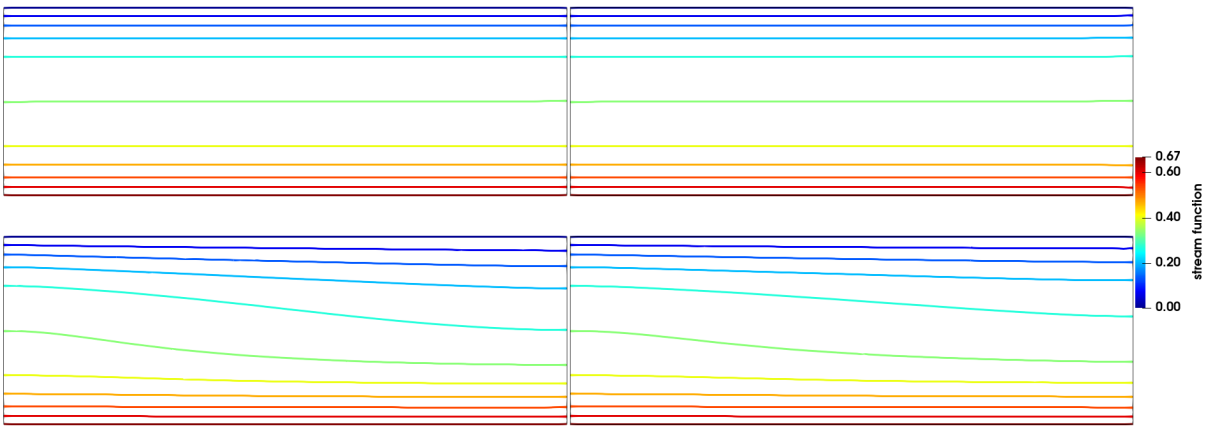


Figure 5: Exercise 3 - streamlines. Left: $G(\mathbf{u}) = \nabla(\mathbf{u})$; right: $G(\mathbf{u}) = \varepsilon(\mathbf{u})$; top: no-slip on Γ_{top} ; bottom: free slip on Γ_{top} .